

CS410 Midterm 3 Part B

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I. QUESTION (25 POINTS)

Design a single tape TM that recognizes the language $L = \{0^n 1^p 2^q \mid p + q \text{ is a multiple of } n\}$. Draw a state diagram, give states, transitions, etc.

II. QUESTION (25 POINTS)

Describe a single tape TM for the following language: $D = \{w \# x \mid w^R \text{ is a substring of } x \text{ for } w, x \in \{0, 1\}^*\}$. Draw a state diagram, give states, transitions, etc.